IEOR 4404: Simulation

Lecture 10: Simulating Markov Chains

Discrete-Time Markov Chains

A discrete-time stochastic process is a sequence of random variables, typically indexed by time:

$$X_0, X_1, X_2, \ldots$$

In contrast, continuous-time: $X_t, t \geq 0$

Markov chains are stochastic processes characterized by "memoryless": Future states only depend on the current state, but not those before it

$$P(X_{n+1} = x_{n+1} \mid X_0 = x_0, \ldots, X_n = x_n) \\ = P(X_{n+1} = x_{n+1} \mid X_n = x_n)$$

Discrete-Time Markov Chains

if it depends on n, then it's not homogeneous.

Time homogeneity:

- If $P(X_{n+1} = x_{n+1} \mid X_n = x_n)$ does not depend on time step n, then the Markov chain is called (time) homogeneous.

Discrete state space:

- States are countable, e.g., {0,1}, {0,1,2,…}, {H,T}, {(1,1),(1,2),… }
- We will denote states generically as $\{0,1,2, \dots\}$ for convenience

Transition probabilities:

- For time homogeneous discrete-state-space Markov chain, we denote $P(X_{n+1} = j \mid X_n = i)$ by p_{ij}, which are called the transition probabilities

Initial distribution:

- X_0 may follow a distribution specified by $P(X_0 = i)$ for i in the state space

Discrete-Time Markov Chains

We can represent the transition probabilities via a transition matrix (supposing states are labeled as 0,1,2…)

$$P = \begin{bmatrix} p_{00} & p_{01} & \cdots \\ p_{10} & p_{11} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Example:

$$\begin{array}{cc} \text{state} & \begin{array}{ccc} 0 & \quad 1 & \quad 2 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} & P = \begin{bmatrix} 0.1 & 0.2 & 0.7 \\ 0.5 & 0 & 0.5 \\ 0.4 & 0.6 & 0 \end{bmatrix} \end{array}$$

$P_{21} = 0.6$

- Here, the probability of transitioning from state 0 to state 1 is $p_{01} = 0.2$, etc.
- Note that the row sums are always 1

Discrete-Time Markov Chains

Quantities of interest:

transient quantities

- Probability or expectation at a future time:

$$P(X_n = j | X_0 = i), E[g(X_n) | X_0 = i], P(X_n = j), E[g(X_n)]$$

- First passage probabilities or expectation:

$$P(N < \infty | X_0 = i) \text{ or } E[N | X_0 = i]$$

where $N = \min\{n: X_n \in A\}$ first hitting time to A.

- Steady-state distribution or expectation:

stationary distribution

$$\lim_{n \to \infty} \frac{\#\{visits\ to\ state\ i\ before\ time\ n\}}{n} \text{ or } P_\pi(X = i)$$

long-run frequency of state i

$$\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} g(X_k) \text{ for a "cost" function } g(\cdot)$$

- A variety of other quantities

Simulating Markov Chains

Simulating Markov chains is easy..

Idea: If current state is i, generate the next state according to transition probabilities p_{ij}

i.e., generate X_T

Suppose we want to simulate the distribution of X_T, given $X_0 = i$. Pseudo code:

- Set $X = i$
- For $n = 0,1,2, \ldots, T$: Given $X = i$, set $X = j$ with probability $p_{i,j}$

Or even more compactly as:

- Set $X = i$
- For $n = 0,1,2, \ldots, T$: Update X to j with probability $p_{X,j}$

Simulating Markov Chains

We can also keep track of the entire trajectory.
Pseudo code:

- Set $n = 0, X_0 = i$
- While (stopping criterion is not met) do:
  - Generate $X_{n+1} = j$ with probability $p_{X_n,j}$
  - $n = n + 1$

Simulating Markov Chains

Sometimes it is easier to define and simulate Markov chain by

$$X_{n+1} = f(X_n, V_{n+1})$$

dist of X_{n+1} depends only on X_n

where f is a deterministic function, and V_{n+1} is a random variable independent of X_n.

Pseudo code:

- Set $n = 0, X_0 = i$
- While (stopping criterion is not met) do:
  - Generate V
  - Set $X_{n+1} = f(X_n, V)$
  - $n = n + 1$

Simulating Markov Chains

In fact, when we use ITM to draw from the transition probabilities, we implicitly use this representation:

$$X_{n+1} = f(X_n, U_{n+1})$$

where $U_{n+1} \sim Unif(0,1)$ and $f(i,u) = j$ if $\sum_{k=0}^{j-1} p_{ik} \leq u \leq \sum_{k=0}^{j} p_{ik}$ with $p_{i,-1}$ defined as 0

Example: Simulate the Markov chain with states 0, 1, 2, and transition matrix

$$P = \begin{bmatrix} 0.1 & 0.2 & 0.7 \\ 0.5 & 0 & 0.5 \\ 0.4 & 0.6 & 0 \end{bmatrix}$$

starting from initial state 0, assuming you only have access to the generation of Unif(0,1).

$u \sim unif(0,1)$

$f(i,u)$ = "inverse cdf" for discrete r.v. with pmf $p_{i0}, p_{i1}, \ldots$

0 $\quad$ 1

$\underbrace{\qquad}_{p_{i0}}$ 0 $\quad$ $\underbrace{\qquad}_{p_{i1}}$ 1 $\quad$ $\underbrace{\qquad}_{p_{i2}}$ 2 $\quad \ldots$

$$\mathbb{P} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.1 & 0.2 & 0.7 \\ 0.5 & 0 & 0.5 \\ 0.4 & 0.6 & 0 \end{bmatrix} \end{array}$$

To generate X_T, starting from $X_0 = 0$

- initiate $X = 0$
- for $n = 1, \ldots, T$: given $X = i$,

update $X = 0$ with prob P_{i0}

$\quad\quad 1 \quad \ldots \quad P_{i1}$

$\quad\quad 2 \quad \ldots \quad P_{i2}$

- initiate $x = 0$
- for $n = 1, \dots, T$: Given $X = i$,

  update $X = f(i, u)$ where $u \sim \text{unif}(0,1)$,

$$f(i,u) = \begin{cases} 0 & \text{if} \quad 0 < u \leq p_{i0} \\ 1 & \text{if} \quad p_{i0} \leq u < p_{i0} + p_{i1} \\ 2 & \text{if} \quad p_{i0} + p_{i1} \leq u < 1 \end{cases}$$

Simulating Markov Chains

The alternative representation for Markov chain

- can be the natural choice in many cases
- can apply beyond discrete state space

interarrival times follow general dist.

1 server

Example: In a first-come-first-serve G/G/1 queue, waiting times are represented via Lindley's recursion:

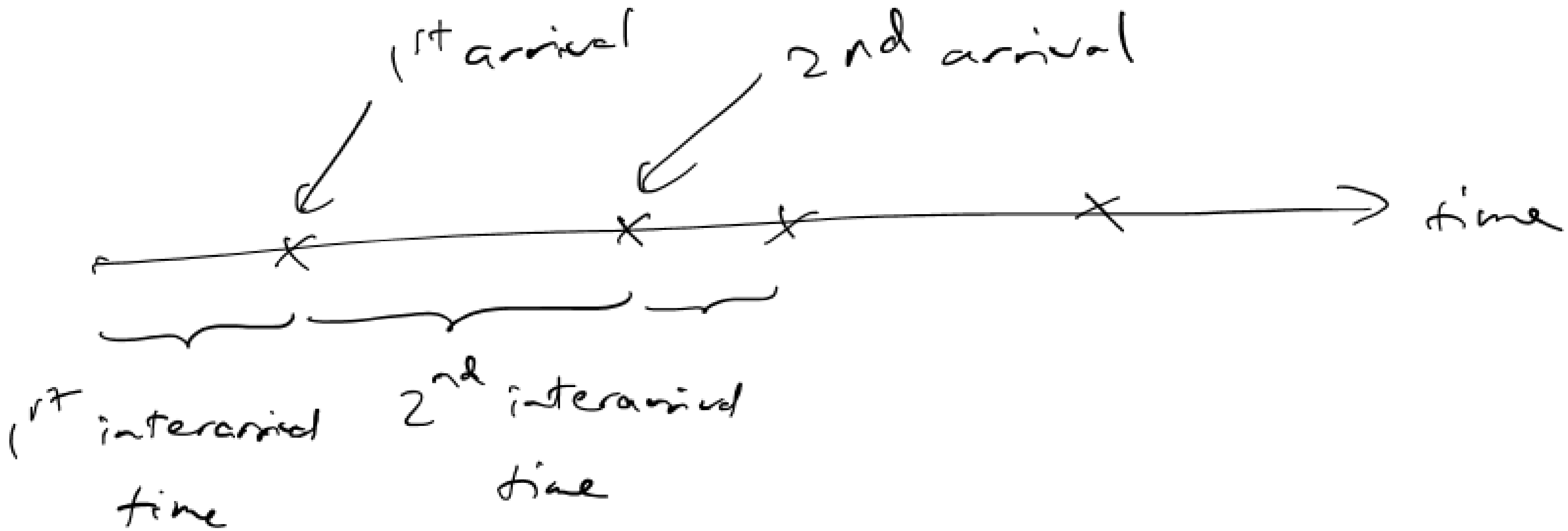
$$W_{n+1} = [W_n + C_n]^+$$

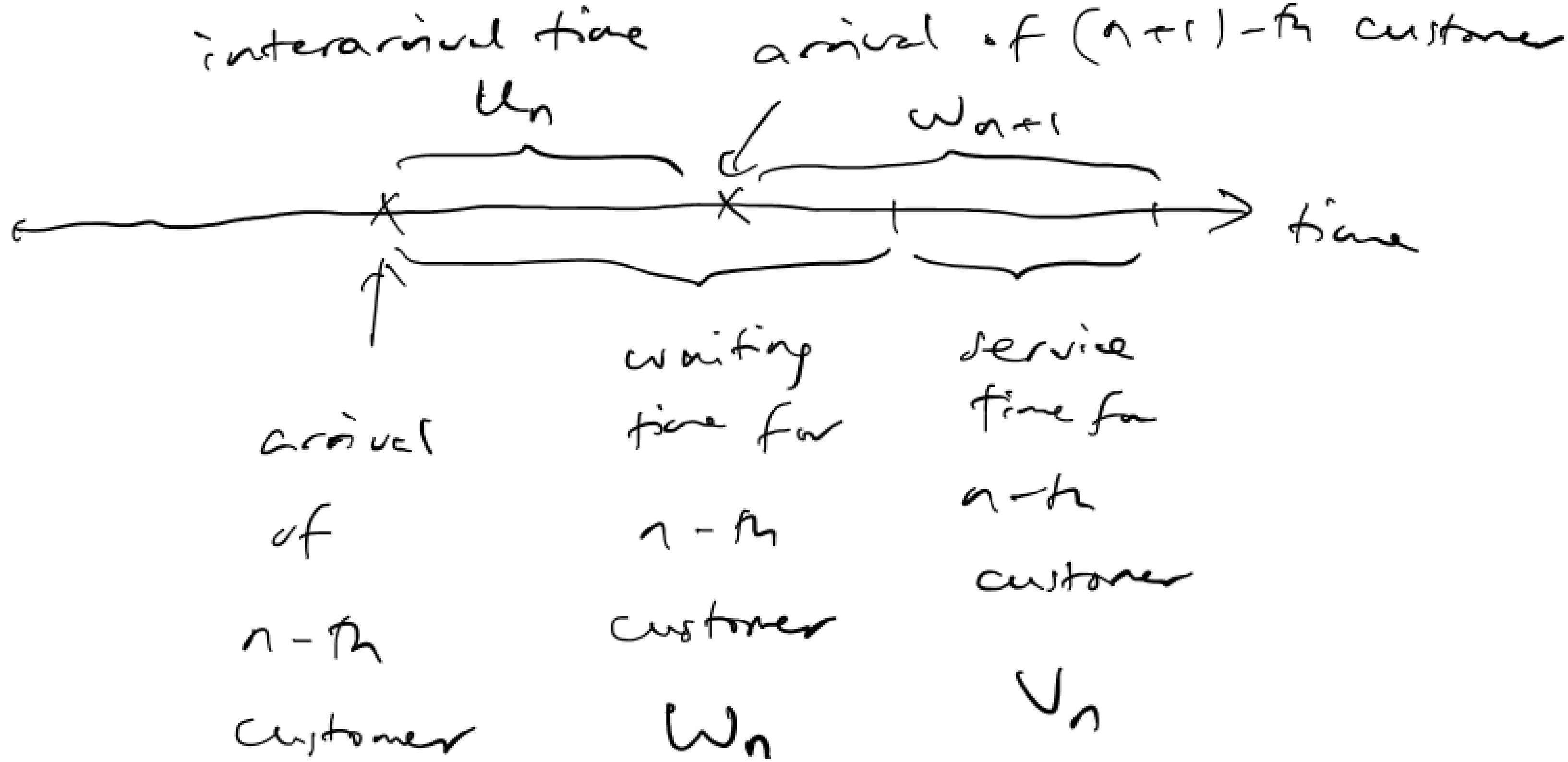
service times follow general dist.

where C_n is a random variable independent of W_n

To simulate it,

- Set $n = 0$, $W_0 = 0$
- While (stopping criterion is not met) do:
  - Generate C_n
  - Set $W_{n+1} = [W_n + C_n]^+$


1st arrival
2nd arrival
time
1st interarrival time
2nd interarrival time


interarrival time
U_n
arrival of $(n+1)$-th customer
W_{n+1}
time
arrival of n-th customer
waiting time for n-th customer
W_n
service time for n-th customer
V_n

$$W_{n+1} = (W_n + V_n - U_n)^+$$

Simulating Markov Chains

Other examples:

- Random walk on integers: $p_{i,i+1} = p,\ p_{i,i-1} = 1 - p$

- Running maximum: Let $Y_1, Y_2, \ldots$ be i.i.d. random variables. Then, $M_n = \max\{Y_1, \ldots, Y_n\}$ form a Markov chain

$$M_{n+1} = \max\{M_n, Y_{n+1}\}$$

$X_{n+1} = X_n + \xi_{n+1}$ where

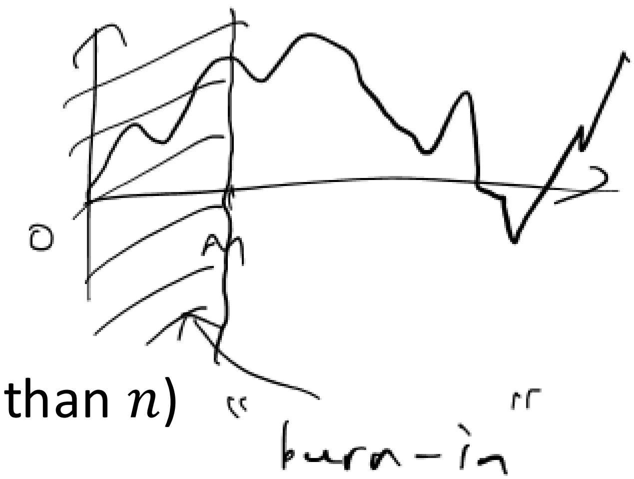
$$\xi_{n+1} = \begin{cases} +1 & \text{w.p. } p \\ -1 & \text{w.p. } 1-p \end{cases}$$

Steady-State Simulation

For steady-state quantities, we simulate the Markov chain for a long time, and remove the beginning part of the chain to alleviate the initial bias

Suppose we want to estimate $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} g(X_k)$. We can simulate $X_k, k = 0, \dots, n$ for a large n. Then output

$$\frac{1}{n-m}\sum_{k=m+1}^{n} g(X_k)$$



where m is a reasonably large number (but less than n)

The first m steps are called the "burn-in" period that is removed from the final calculation. Determining the length of burn-in is case-by-case (depending on the mixing time of the Markov chain).

Simulating Markov Chains

Connection to the theory of Markov chains:

In theory,

- Calculating the n-th step transition uses the Chapman-Kolmogorov equations and matrix multiplications
- Calculating the steady-state distribution uses the stationary equations or, in the case the Markov chain is time reversible, the detailed balance equations

In simulation, both problems can be approximated routinely